Research Notes on Recommender system

Recommender system

>> Section 1

Generative recommenders Generative recommenders (GR) represent an approach that transforms recommendation tasks into sequential transduction problems, where user actions are treated like tokens in a generative modeling framework. In one method, known as HSTU (Hierarchical Sequential Transduction Units), high-cardinality, non-stationary, and streaming datasets are efficiently processed as sequences, enabling the model to learn from trillions of parameters and to handle user action histories orders of magnitude longer than before. By turning all of the system’s varied data into a single stream of tokens and using a custom self-attention approach instead of traditional neural network layers, generative recommenders make the model much simpler and less memory-hungry. As a result, it can improve recommendation quality in test simulations and in real-world tests, while being faster than previous Transformer-based systems when handling long lists of user actions. Ultimately, this approach allows the model’s performance to grow steadily as more computing power is used, laying a foundation for efficient and scalable “foundation models” for recommendations.

>> Section 2

Reinforcement learning for recommender systems The recommendation problem can be seen as a special instance of a reinforcement learning problem whereby the user is the environment upon which the agent, the recommendation system acts upon in order to receive a reward, for instance, a click or engagement by the user. One aspect of reinforcement learning that is of particular use in the area of recommender systems is the fact that the models or policies can be learned by providing a reward to the recommendation agent. This is in contrast to traditional learning techniques which rely on supervised learning approaches that are less flexible, reinforcement learning recommendation techniques allow to potentially train models that can be optimized directly on metrics of engagement, and user interest.

>> Section 3

Weighted: Combining the score of different recommendation components numerically. Switching: Choosing among recommendation components and applying the selected one. Mixed: Recommendations from different recommenders are presented together to give the recommendation. Cascade: Recommenders are given strict priority, with the lower priority ones breaking ties in the scoring of the higher ones. Meta-level: One recommendation technique is applied and produces some sort of model, which is then the input used by the next technique.

>> Section 4

Recommend systems are essential for modern e-commerce platforms, playing a key role in improving the customer experience and increasing sales. These systems analyze customer data to provide personalized product suggestions, helping users discover items they might not have found on their own. A study by J. Leskovec et al. highlighted that such systems are crucial when an individual needs to choose from a potentially overwhelming number of items that a service may offer. E-commerce recommenders typically use a combination of filtering techniques to generate these suggestions. Collaborative filtering is a core method, recommending products based on the purchasing and Browse habits of similar users. Another widely used approach is content-based filtering, which recommends items with similar attributes to those a user has previously shown interest in. Many e-commerce platforms use a hybrid approach, combining these techniques to create more accurate and diverse recommendations, which helps to address issues like the "cold start" problem for new users or products. These systems are implemented in several ways across e-commerce sites to maximize their effectiveness at different stages of the shopping process:

>> Section 5

Last.fm creates a "station" of recommended songs by observing what bands and individual tracks the user has listened to on a regular basis and comparing those against the listening behavior of other users. Last.fm will play tracks that do not appear in the user's library, but are often played by other users with similar interests. As this approach leverages the behavior of users, it is an example of a collaborative filtering technique. Pandora uses the properties of a song or artist (a subset of the 450 attributes provided by the Music Genome Project) to seed a "station" that plays music with similar properties. User feedback is used to refine the station's results, deemphasizing certain attributes when a user "dislikes" a particular song and emphasizing other attributes when a user "likes" a song. This is an example of a content-based approach. In e-commerce, Amazon's well-known "customers who bought X also bought Y" feature is a prime example of collaborative filtering. It also uses content-based filtering when it recommends a book by the same author you've previously read or a pair of shoes in a similar style to ones you've viewed. Each type of system has its strengths and weaknesses. In the above example, Last.fm requires a large amount of information about a user to make accurate recommendations. This is an example of the cold start problem, and is common in collaborative filtering systems.

Whereas Pandora needs very little information to start, it is far more limited in scope (for example, it can only make recommendations that are similar to the original seed).

>> Section 6

Data Representation: Create a n-dimensional space where each axis represents a user's trait (ratings, purchases, etc.). Represent the user as a point in that space. Statistical Distance: 'Distance' measures how far apart users are in this space. See statistical distance for computational details Identifying Neighbors: Based on the computed distances, find k nearest neighbors of the user to which we want to make recommendations Forming Predictive Recommendations: The system will analyze the similar preference of the k neighbors. The system will make recommendations based on that similarity

>> Section 7

Cold start: For a new user or item, there is not enough data to make accurate recommendations. Note: one commonly implemented solution to this problem is the multi-armed bandit algorithm. Scalability: There are millions of users and products in many of the environments in which these systems make recommendations. Thus, a large amount of computation power is often necessary to calculate recommendations. Sparsity: The number of items sold on major e-commerce sites is extremely large. The most active users will only have rated a small subset of the overall database. Thus, even the most popular items have very few ratings. One of the most famous examples of collaborative filtering is item-to-item collaborative filtering (people who buy x also buy y), an algorithm popularized by Amazon.com's recommender system. Many social networks originally used collaborative filtering to recommend new friends, groups, and other social connections by examining the network of connections between a user and their friends. Collaborative filtering is still used as part of hybrid systems. This technique can employ embeddings, a machine learning technique.